

HEEYEON KOO

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SUMMARY

Aspiring AI engineer specializing in large language models, PyTorch, and scalable ML pipelines. Experienced in data preprocessing, model evaluation, and fine-tuning; Co-authored research published in AAAI and ADVEI Journal.

EDUCATION

University of Texas, Dallas

Master of Data Science and Statistics / Concentration: Applied Statistics

Richardson, Texas

August. 2025 – Present

- NSM McDermott Master's Student Fellowship (Fall 2025)

Yonsei University

Master of Artificial Intelligence (GPA: 4.15 / 4.5)

Seoul, Korea

Sept. 2023 – August. 2025

- Teaching Assistant, "Computer Programming" (Fall 2024)
- Admission Scholarship (Fall 2023)

Kyung Hee University

Bachelor of Computer Science and Engineering (GPA: 3.91 / 4.5)

Gyeonggi-Do, Korea

Mar. 2018 – Feb. 2023

- Academic Excellence Scholarship (Spring 2018); Software Scholarship (Spring 2022)

EXPERIENCE

Yonsei University

Seoul, Korea

Graduate Research Student | Internet Computing Laboratory

Aug. 2023 – Aug. 2025

- Collaborated with a 4-member team and Hyundai Motor Company to architect an ontology-based information extraction framework by fine-tuning llama-based LLMs with a domain-specific automotive knowledge graph.
- Designed and implemented annotation guidelines for 422 automotive documents, constructing a knowledge graph with 7 entity classes and 22 relation types, achieving 93.7% coverage across 684 competency questions.
- Fine-tuned LLMs using the constructed KG and evaluated 5 event argument extraction baselines, achieving a +66% F1 score improvement over state-of-the-art models.
- Developed a retrieval-augmented generation (RAG)-based complex claim fact-checking framework leveraging advanced prompt engineering using GPT-3.5-turbo, outperforming NLI fine-tuned baselines by +7.92% F1, contributing to a research publication (under ESWA journal review) and an ongoing patent filing.
- Preprocessed and annotated 500 automotive recall records from the NHTSA dataset on Kaggle for NER and relation tagging, evaluating precision, recall, and F1 across diverse manufacturing document sets.

Seoul National University

Seoul, Korea

Research Intern | Bio-Intelligence Laboratory

Jan. 2023 – Apr. 2023

- Assisted in drafting methodology and results sections for a representation learning paper using LaTeX.
- Organized and synthesized research notes, reviewed recommended publications, and engaged in discussions with the PI to deepen understanding of representation learning.

Electronics and Telecommunications Research Institute

Daejeon, Korea

Research Intern | Media Research Division

Jul. 2022 – Aug. 2022

- Analyzed hologram compression algorithms (interference codecs) and evaluated their performance.
- Applied internship knowledge to a capstone project on phase unwrapping, comparing RMSE performance of Res-UNet, PUMA, and DCT methods for hologram phase compression.

Myzzym

Gyeonggi-do, Korea

Software Engineering Intern | Software Engineering Division

Dec. 2020 – Feb. 2021

- Built a responsive storage platform using React.js, enhancing user experience across multiple devices.
- Created and executed 128 test scenarios for user and admin modes, ensuring cross-browser functionality on IE, Chrome, Microsoft Edge, and Safari.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (React, Nodejs), Java, MATLAB, HTML/CSS, SQL

Libraries: PyTorch, scikit-learn, Spacy, HuggingFace, Transformers, Pandas, NumPy, CUDA

Develop Tools: VS Code, Visual Studio, R Studio, Jupyter Notebook, Android Studio, Overleaf (LaTeX), Git